



The Chemistry a Culture Performs

Student copy (project or hand out)

1 A crock of cabbage sits on a kitchen counter, salted, weighted, sealed away from the air. Nothing about it looks busy. Yet inside, a population that cannot be seen is already at work, breaking glucose without oxygen and converting it, molecule by molecule, into the acid that will keep the harvest from rotting. The cabbage does not change because the cook willed it to. It changes because something living is metabolizing inside it.

2 What that population is doing has a name and a chemistry. Fermentation is anaerobic respiration: microorganisms pull energy from glucose with no oxygen present, and they leave behind organic end products, lactic acid in one pathway, ethanol and carbon dioxide in another. The bacterium most responsible for the souring crock, *Lactobacillus*, runs the first route. One molecule of glucose becomes two molecules of lactic acid, by way of glycolysis and an enzyme called lactate dehydrogenase.

3 That enzyme does not improvise. It catalyzes one reaction, the reduction of pyruvate to lactate, accepting electrons from a carrier called NADH, and in doing so it hands back the NAD^+ that glycolysis needs to keep running; without that handoff the whole sequence would stall after a single pass. The precision is almost hard to credit: lactate dehydrogenase places its atoms with an estimated error rate below one in a billion, so only one optical form of the acid, the D or the L, comes out, depending on which version of the enzyme the microbe carries.

4 Yeast performs a different chemistry on the same sugar. *Saccharomyces cerevisiae*, the organism in bread and beer, splits its work across two enzymes: pyruvate decarboxylase strips a carbon away as carbon dioxide and leaves acetaldehyde, then alcohol dehydrogenase reduces that acetaldehyde to ethanol. The first enzyme needs a scrap of vitamin B1 and a magnesium ion to function. The second holds a zinc atom at its core and passes a hydride from NADH onto each molecule it converts. Both steps exist to do what the bacterium's single step also did: regenerate NAD^+ , so glycolysis can keep yielding its meager two molecules of ATP per glucose.

5 The conditions for all this are narrow, and the cook controls them more than the microbe does. Lactic acid bacteria grow best between 30 and 40 degrees Celsius, in a mild acidity that drifts as they work. Yeast prefers it cooler, 25 to 30 degrees, and turns sluggish and stressed near 35, its ethanol yield falling off. A few degrees is not a detail. It is the difference between the culture a cook wanted and a different one entirely.

6 A starter culture is the one deliberate intervention. Drop a defined population of selected strains into the substrate and they outcompete whatever wandered in, claim the sugar first, drop the pH, and in some cases secrete bacteriocins, small antimicrobial peptides that suppress the organisms a cook does not want. What

follows is not static. The community shifts in sequence: one species dominates, alters the chemistry, and yields the lead to the next as the acid climbs; and in those dense crowded populations, cell pressed against cell, reactions occur that never occur in a microbe kept alone.

7 None of it is simple, and the same culture is not locked to one product. Lactic acid bacteria can route pyruvate several ways depending on what surrounds them, toward lactate, toward acetyl-CoA and carbon dioxide, toward compounds that register on the tongue as flavor. The leuconostocs and weissellas take yet another path, the phosphoketolase route, and yield lactic acid, ethanol, and gas all at once. A cook salts a vegetable and waits. The cook supplies the temperature, the salt, the time, and the harvest. Everything else, the part that actually does the transformation, belongs to something with no intention at all, only a chemistry it cannot help but perform.

AP-style multiple choice:

1. In the fourth paragraph, the writer details yeast's two-enzyme pathway and then notes that both steps exist to regenerate NAD⁺ "so glycolysis can keep yielding its meager two molecules of ATP per glucose." This detail primarily serves to
 - A) argue that ethanol fermentation is more efficient than lactic acid fermentation
 - B) clarify that yeast and bacteria, despite different end products, perform the same underlying chemical service
 - C) establish that yeast requires more cofactors than bacteria and is therefore more advanced
 - D) suggest that the energy yield of fermentation is too small to matter biologically
2. Which of the following best characterizes the writer's tone toward the microorganisms across the opener?
 - A) celebratory enthusiasm that foregrounds the writer's own wonder
 - B) cautious alarm at the risks of uncontrolled microbial activity
 - C) detached skepticism about claims made for fermentation
 - D) measured admiration grounded in technical precision rather than overt praise

Discussion questions:

1. The opener repeatedly frames the cook and the microbe as partners with divided labor: the cook supplies salt, temperature, and time, while the microbe supplies the chemistry. Where in the text is this division clearest, and how does the writer keep it from tipping into either pure human control or pure microbial autonomy?
2. The Long Look register builds wonder through accumulated precise detail rather than through words like astonishing. Find two moments where a technical particular (a number, an enzyme, a temperature) carries more awe than an adjective could, and explain how the precision itself does the emotional work.

Writing prompt (Homework or extended in-class, Q2 rhetorical analysis):

The opener describes fermentation as a process that happens out of sight, framing the relationship between a cook and the living culture that transforms a harvest. Read the passage carefully. Then write an essay in which you analyze the rhetorical choices the writer makes to convey wonder at the chemistry of fermentation while withholding overt expressions of awe. Support your analysis with specific evidence from the text, and explain how the writer's choices of detail, sentence structure, and figurative language create the intended effect on the reader.